

NCERT CLASS 12 PHYSICS: ELECTROMAGNETIC INDUCTION

Section: Magnetic Flux and Faraday Law

Magnetic flux through a surface of area A placed in a uniform magnetic field B is defined as $\Phi = B \cdot A \cdot \cos(\theta)$. The SI unit of magnetic flux is Weber (Wb).

Faraday stated two laws of electromagnetic induction:

1. Whenever magnetic flux linked with an electric circuit changes, an electromotive force (EMF) is induced in the circuit.
2. The magnitude of the induced EMF is directly proportional to the time rate of change of magnetic flux through the circuit: $\epsilon = -\frac{d\Phi}{dt}$.

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Section: Lenz Law and Conservation of Energy

Lenz law states that the polarity of induced EMF is such that it tends to produce a current which opposes the change in magnetic flux that produces it.

The negative sign in Faraday formula $\epsilon = -d\Phi/dt$ represents Lenz law.

Lenz law is a direct consequence of the Law of Conservation of Energy. If the induced current were in the same direction as the change in flux, a small push to a magnet would create infinite energy without external work, which violates physical laws.

NCERT CLASS 12 PHYSICS: ELECTROMAGNETIC INDUCTION

Section: Self Induction and Mutual Inductance

When electric current in a coil changes, an induced EMF is produced in the same coil. This phenomenon is called Self-Induction: $e = -L \cdot (di/dt)$, where L is the self-inductance measured in Henry (H).

Mutual induction occurs between two adjacent coils: $e_2 = -M \cdot (di_1/dt)$, where M is the mutual inductance coefficient.